Reg. No.	9	5	1	7									

Question Paper Code : 21057

B.E. / B.TECH. DEGREE EXAMINATIONS, MAY / JUNE 2024

Fourth Semester

Artificial Intelligence and Data Science

19AD403 – PROGRAMMING WITH JAVA

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

U, CO1 1. Identify the correct restriction on static methods.

1. They must access only static data.
2. They can only call other static methods.
3. They cannot refer to this or super.

A) 1 and 2 B) 2 and 3 C) Only 3 D) 1, 2 and 3

R, CO1 2. Write Java code to create an abstract class and show how the abstract class is getting used?

A, CO2 3. Write a Java program to create ArrayList to store 5 random integer values. Then perform deletion of the third element in the array list, then sort the numbers in the Array List.

U, CO2 4. What will be the output of the following code snippet? Justify your answer.

```
try{
    int a=10;
    int c=a/0;
}
catch (Exception e)
{
    System.out.println("in Exception");
}
catch (ArithmeticException e)
{
    System.out.println("in Arithmetic Exception");
}
```

A) In Exception

B) In Arithmetic Exception

C) In Exception
In Arithmetic Exception

D) Unreachable catch block for Arithmetic Exception

A, CO3 5. class one

```
{
    int a = 10;
}
```

The object of the above ***class one*** needs to be serialized in a file called abc.txt

Write a program in Java to achieve the objective.

U, CO3 6. What is invalid about serialization in Java? Justify your answer.

A) Serialized object can be persisted into database

B) Serialized object can't be persisted into file

C) Serialization is process of converting object into byte stream

D) Serialized object can be transferred over network

A, CO4 7. Write a simple Java code to Create a button which reads "Submit" and make the button invisible and put the button in frame.

U, CO4 8. Which event listener interface is used for handling item selection events in a JComboBox? Justify your answer.

A) ActionListener B) ItemListener

C) MouseListener D) KeyListener

A, CO5 9. Give out a simple Java code to create and run a thread using *Runnable* interface.

U, CO5 10. How can you achieve synchronization between threads in Java? Justify your answer.

A) By using interrupt() methods

B) By using wait() and notify() methods

C) By using multiple catch blocks

D) By using synchronized keyword or synchronized blocks

PART B – (5 × 16 = 80 Marks)

A, CO1 11. a) i. Create a class called Vehicle and show the demonstration of overloading with Java code for the method named calculateMileage() which will be calculated with

A) age of vehicle alone (mileage = 2 / age)

B) kms travelled and age of vehicle (mileage = 2 / (age + kms)

C) cylinder volume (mileage = 3/ colume)

D) cylinder volume and number of cylinders (mileage = 3/ (volume * no. of cylinders) (10 Marks)

U, CO1 11. a) ii. What is the significance of having static method, static variable, final method, final variable? Give suitable Java code. (6 Marks)

OR

A, CO1 11. b) i. Write Java code for the following scenario:

In an application development, you are supposed to develop an abstract class named *abc* having abstract function fun1(), and a normal function named fun2(). Class named *xyz* and *mno* are going to use the abstract class abc and implement its abstract methods. Now you have one more class called *rrr* which inherits *xyz* and show the overriding of fun1 method. (10 Marks)

U, CO1 11. b) ii. Show the following scenarios using Java codes: (Do not write any body for the methods)

A) Interface extending many interfaces

B) Declare integer variables in interface and use the variables in classes which implements the interface

C) Multilevel inheritance for two levels.

D) Class implementing many interfaces and inheriting one class. (6 Marks)

A, CO2 12. a) i. Use Java Exception handling mechanism for the following scenarios. Give Java code to show how the control flow will be for the following scenarios: *(Note: write code after the try/catch/finally block)*

Scenario	Exception occurred	Catch block	Finally block
1	Yes	Yes	Yes
2	Yes	No	Yes
3	Yes	Yes	No
4	No	Yes	Yes

(10 Marks)

U, CO2 12. a) ii. Define exception propagation with an example code and show exception handling with multiple catch blocks. (6 Marks)

OR

A, CO2 12. b) i. Write a program in Java to convert infix to postfix expression. Use Java collection named 'Stack' to achieve the objective. *(Note: keep the operators such as +, - , *, / , ^ in the infix expression)* (10 Marks)

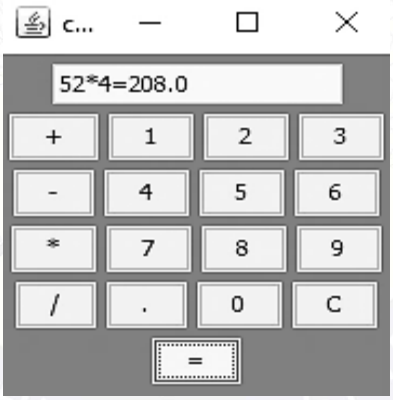
U, CO2 12. b) ii. How can you create a Priority Queue which stores strings with descending priority? *(Note: meaning "xyz" is having more priority than "xar")* (6 Marks)

U, CO3 13. a) Explain in detail about Byte steam and Character stream. Demonstrate byte steam and character steam based reading and writing of data. (16 Marks)

OR

U, CO3 13. b) Define Serialization and Deserializations and its needs. Explain it with a sample code. (16 Marks)

A, CO4 14. a) Write a program in Java using Java swing to create calculator application. Stick to basic arithmetic operations for the calculator. *(Create a calculator similar to the below one with little adjustments)*



(16 Marks)

OR

A, CO4 14. b) Write a program in Java using java swing for two player tic tac toe game. You can develop UI based on given design.

(16 Marks)

U, CO5 15. a) i. Explain the following thread related items with a necessary simple Java code:

A) Significance of join() method

B) The purpose and working of sleep() method.

C) Daemon threads (10 Marks)

U, CO5 15. a) ii. Define the following in detail with an example code wherever its necessary:

A) States in the lifecycle of a thread

B) Atomic action in Java (6 Marks)

OR

U, CO5 15. b) i. Explain the following thread related items with necessary simple Java code:

A) Is it possible to start a thread twice?

B) What are synchronized blocks, and what is their purpose?

C) What is the function of the Yield method in a Thread-class? (10 Marks)

U, CO5 15. b) ii. Define the following in detail with an example code wherever its necessary:

A) Differentiate between the Thread class and Runnable interface for creating a Thread.

B) What are some fundamental advantages of multithreading in Java? (6 Marks)